INCH-POUND

MIL-T-82796(OS)
29 December 1989
SUPERSEDING
WS 20639
25 October 1983

MILITARY SPECIFICATION

TRIMESOYL 1-(2-ETHYL) AZIRIDINE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the requirements for tremesoyl 1-(2-ethyl) aziridine, a butylene imine derivative of trimesic acid, referred to herein as BITA.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202640-23T07:26Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specificatins and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 203 Water Using Karl Fischer Reagent (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **Chemical and physical properties.** The chemical and physical properties of BITA shall conform to table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Imine equivalent weight	135	150
Moisture content, weight percent	---	0.25
Hydrolyzable chloride, weight percent	---	0.9

3.2 **Safety.**

3.2.1 **Material Safety Data Sheets (MSDS).** The contractor shall prepare and submit an MSDS in accordance with FED-STD-313 as specified in the contract (see 6.2).

3.2.2 **Toxic products.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.3 **Shelf life.** The BITA specified herein shall have a shelf life of 12 months after the date of acceptance testing when stored at 0°F maximum.

3.3.1 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for additional 12-month periods provided the BITA, upon retest, conforms to the imine equivalent weight and moisture content requirements specified in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

3.3.2 **Allowable temperature excursions during storage.** To allow 80 ± 20°F material conditioning prior to use, a 336-hour maximum cumulative conditioning time above 0°F shall be acceptable during the 12-month material storage life. BITA that has exceeded 336 hours exposure to temperatures above 0°F may have the storage life extended as specified in 3.3.1.

NOTE: The requirements specified in 3.3, 3.3.1, and 3.3.2 apply only to the procuring activity.

3.4 **Workmanship.** Workmanship shall be such that BITA, when sufficiently warmed to become liquid, is uniform in appearance and free from visible contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the tests specified in table II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

TABLE II. Quality conformance inspection.

Examinations and Tests	Requirement	Test Method
Visual examination	3.4 and Section 5	4.3.1
Imine equivalent weight	3.1	4.3.2
Moisture content	3.1	4.3.3
Hydrolyzable chloride	3.1	4.3.4

4.2.1 **Lot.** A lot shall consist of all the material subject to inspection at the same time and manufactured by identical processes from one of the following:

a. A single batch (see 6.3.1)
b. A continuous process (see 6.3.2)
c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.2.2 **Sampling.** A representative sample of sufficient size to perform all tests and retain a sample shall be taken at approximately the midpoint of packaging each lot. The number of containers selected shall be in accordance with table III. The single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of fewer than three containers, three samples shall be taken from each container. The sample portions thus obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table I. Failure of any composite sample to meet any of the requirements shall be cause for rejection of the lot.

TABLE III. Sampling.

Number of containers in lot	Number of containers to be selected
1 - 3	all
4 - 40	3
41 - 65	4
66 - 110	5
111 - 180	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

4.3 **Inspection methods.** The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.3.1 **Visual examination.** Visual examination shall be made to verify conformance to the requirements of 3.4 and section 5.

4.3.2 **Imine equivalent weight procedures.** The imine equivalent weight shall be determined in accordance with the following:

a. Run in duplicate. Add 50 ml of methanolic potassium thiocyanate (KSCN) solution (prepared by dissolving 160 grams of KSCN in methanol and diluting to 1 liter) into a 125-milliliter Erlenmeyer flask fitted with a nitrogen purge.

b. Pipet 10 ml of 0.4N methanolic hydrochloric acid (prepared by thoroughly mixing 34 ml of concentrated hydrochloric acid in methanol and diluting to 1 liter) into the flask, and thoroughly mix the two solutions.

c. Weigh, to the nearest 0.1 milligram, 0.3 to 0.4 g of sample into the flask, and stir for 5 to 6 minutes.

d. Add three drops each of 0.1-percent alcoholic bromothymol blue sodium salt and 0.1-percent alcoholic phenol red sodium salt indicators. Titrate to a purple endpoint with standardized 0.1N alcoholic potassium hydroxide (KOH) (prepared by dissolving 6 g of KOH in methanol and diluting to 1 liter).

e. Determine two reagent blanks by the above method.

f. Calculate equivalent weight as follows:

$$\text{Equivalent weight} = \frac{W \times 1000}{(B - A)\,N}$$

Where:

A = volume of alcoholic KOH used for sample, ml
B = volume of alcoholic KOH used for blank, ml
N = normality of alcoholic KOH
W = weight of sample, g

g. Record results to the nearest whole unit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

4.3.3 **Moisture content.** Moisture content shall be determined in accordance with ASTM E 203 using a 5.0 ± 0.1-gram sample titrated to a 30-second endpoint and using a solvent of methanol:chloroform, 1:1 by volume. Moisture content shall be recorded to the nearest 0.01 percent.

4.3.4 **Hydrolyzable chloride content.** The hydrolyzable chloride content shall be determined in accordance with the following:

a. Run in duplicate. Weigh, to the nearest 0.1 mg, a 1-g sample into a 250-ml flask with a ground glass neck.

b. Add 50 ml of 0.5N alcoholic potassium hydroxide (prepared by dissolving 30 g of potassium hydroxide in methanol and diluting to 1 liter) and a few boiling chips to the flask.

c. Insert a condenser into the flask and heat to maintain a steady drop rate for 45 minutes (start timing when first reflux begins).

d. Allow the solution to cool, rinse the condenser with approximately 50 ml methanol, and disconnect the flask.

e. Quantitatively transfer the solution to a 400-ml beaker, using small portions of methanol.

f. Using methyl red indicator, neutralize the solution by dropwise addition of concentrated nitric acid solution, then add 3 ml excess.

g. Titrate potentiometrically with standardized 0.02N silver nitrate (prepared by dissolving 3.4 g silver nitrate in distilled water and diluting to 1 liter) using a combination silver-billet/double junction glass electrode and an automatic titrator. Run a reagent blank using the same method.

h. Calculations:

$$\text{Percent hydrolyzable chloride} = \frac{3.545\ (V - B)\ N}{W}$$

Where:

V = volume of standardized 0.02N silver nitrate used for sample, ml

B = volume of standardized 0.02N silver nitrate used for blank, ml

N = normality of standardized 0.02N silver nitrate

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

W = weight of sample, g

3.545 = 1/10 atomic weight of chloride

i. Record hydrolyzable chloride to the nearest 0.1 percent.

4.6 **Inspection of packaging.** The BITA containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein. The BITA shall be packed in dry ice (solid carbon dioxide) in an insulated container and transported as specified in 5.1.1.

5.1.1 **Level C.** The BITA shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.3. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or purchase order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Date of manufacture.

d. Net weight of contents.

e. Shelf life and inspection/test date for extension (see 3.3.1). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTHS PROVIDED THE BITA, UPON RETEST, CONFORMS TO THE IMINE EQUIVALENT AND MOISTURE CONTENT REQUIREMENTS OF MIL-T-82796."

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

f. Storage conditions

g. Toxicity warning (if applicable)

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The BITA is intended for use in the manufacture of Standard Missile rocket motor liner material.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

c. Material Safety Data Sheets Required (see 3.2 and 6.5).

d. Special marking, if other than as specified in 5.2.2.

e. Safety precautions required (see 6.6).

6.3 **Definitions.**

6.3.1 **Batch.** A batch is that amount of BITA made as one unit in the manufacturing process.

6.3.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the BITA is accumulated for the lot.

6.4 **Possible material.** The following material has been found to meet the requirments of this specification. This is only given for information and is not restrictive. HX-868 manufactured by 3M Corporation, St. Paul, Minnesota, has been found to meet the requirements of this specification.

6.5 **Material Safety Data Sheets.** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) Part 23, subpart 23.3. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

6.6 **Safety precautions.** The safety precaution requirements of the "Contractor's Safety Manual for Ammunition, Explosives, and Related Dangerous Material" (DOD 4145.26M) are applicable and should be specified in the contract as required by the DOD FAR Supplement 28.7102.

NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety precaution requirements of "Ammunition and Explosives Ashore" (OP5) should be made applicable.

6.7 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

Preparing Activity:
NAVY-OS
(Project 6810-N051)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Tank-Automotive Command
ATTN: AMSTA-GDS
Warren, MI 48397-5000

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82796	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
TRIMESOYL 1-(2-ETHYL) AZIRIDINE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 20261023107:26Z
Check the source to verify that this is the current version before use.